RiskPulse AI

From Unstructured Financial Signals to Explainable Portfolio Risk Intelligence

Text-to-Risk Signals → Simulated Rebalancing → Synthetic Portfolio Stress Testing

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Problem, Approach & Innovation

PROBLEM

Financial text streams are high-volume and noisy

THE GAP

Sentiment alone does not reveal portfolio vulnerability

OUR ANSWER

An integrated, simulation-driven decision-support approach

RISKPULSE AI

integrated decision-support workflow

1 → Normalize text into traceable signals

2 → Classify events and estimate sentiment and impact where implemented

3 → Connect signals to simulated allocation

4 → Run scenario analysis to inform risk-aware decisions

BASIC SENTIMENT-ONLY

typical workflow

1 → Detect sentiment from text

2 → Generate sentiment score or label

3 → Surface sentiment summary

Stops at sentiment; no link to portfolio vulnerability or scenarios

Feature-level comparison of workflow coverage. RiskPulse AI is an educational decision-support prototype.

System Architecture & Data Flow



NLP Financial Risk Engine

From financial text to decision-ready signals — only where supported.

1. Ingest

Financial text streams

→

2. Validate & Normalize

Clean and standardize for consistency

→

3. Extract & Deduplicate

Entity and event extraction with deduplication

→

4. Model & Score

FinBERT when supported
Deterministic fallback when not

→

5. Signals

Structured signals for downstream decision support

Illustrative synthetic signal example

INPUT TEXT

Company misses quarterly revenue guidance due to supply chain disruption.

→

SENTIMENT	IMPACT	CONFIDENCE	MODEL PROVENANCE
-0.62	7.1	0.78	FinBERT with deterministic fallback enabled
range: -1 to +1	range: 1 to 10	range: 0 to 1	

Illustrative example only. Output shown only where supported.

RiskPulse AI is an educational decision-support prototype, not financial advice or autonomous trading.

Downstream Analytics

1 SIMULATED TACTICAL INDEX REBALANCING

Compares original vs proposed weights

Explains signal rationale, if implemented

Turnover = 0.5 × sum of absolute weight changes

OUTPUT: PENDING PROTOTYPE VERIFICATION



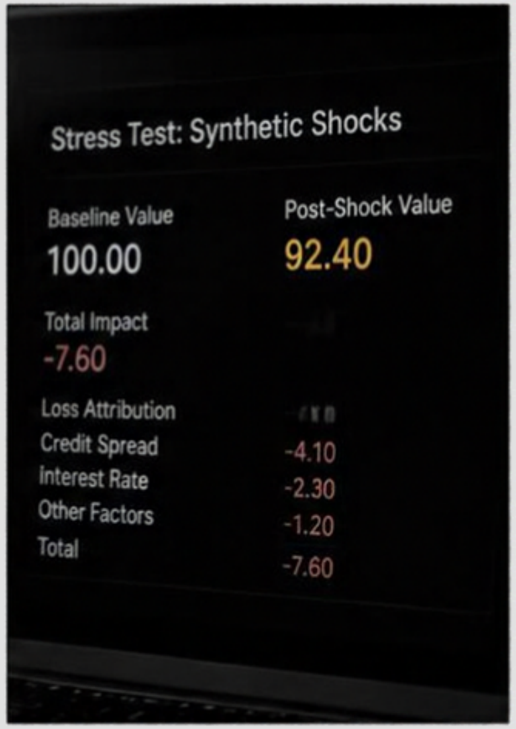
2 SYNTHETIC PORTFOLIO STRESS TESTING

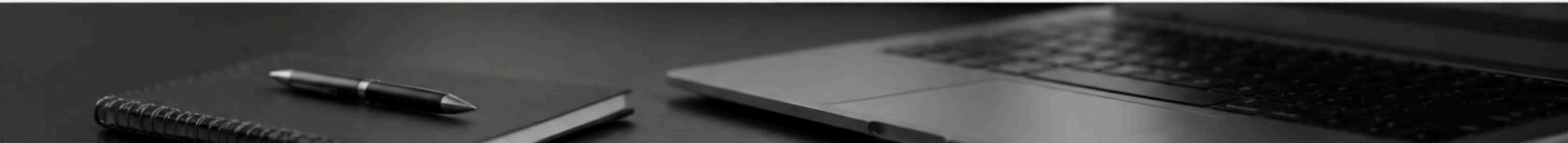
Applies synthetic shocks to wholesale-banking portfolio

Reports before/after value and loss attribution

ECL or duration included, if implemented

OUTPUT: PENDING PROTOTYPE VERIFICATION





Prototype Evidence, Differentiation & Domain Impact

EVIDENCE & TRANSPARENCY

Reserve space for dashboard screenshots and real input-to-output results

Metrics not yet measured

Placeholders until evidence is captured

WHAT SETS RISKPULSE AI APART

Integrated text intelligence plus portfolio analytics

Explainability and source provenance where supported

Tactical and scenario-analysis horizons

VALUE FOR RISK ANALYSTS

Faster signal discovery from noisy financial text

Clearer context for risk assessment and monitoring

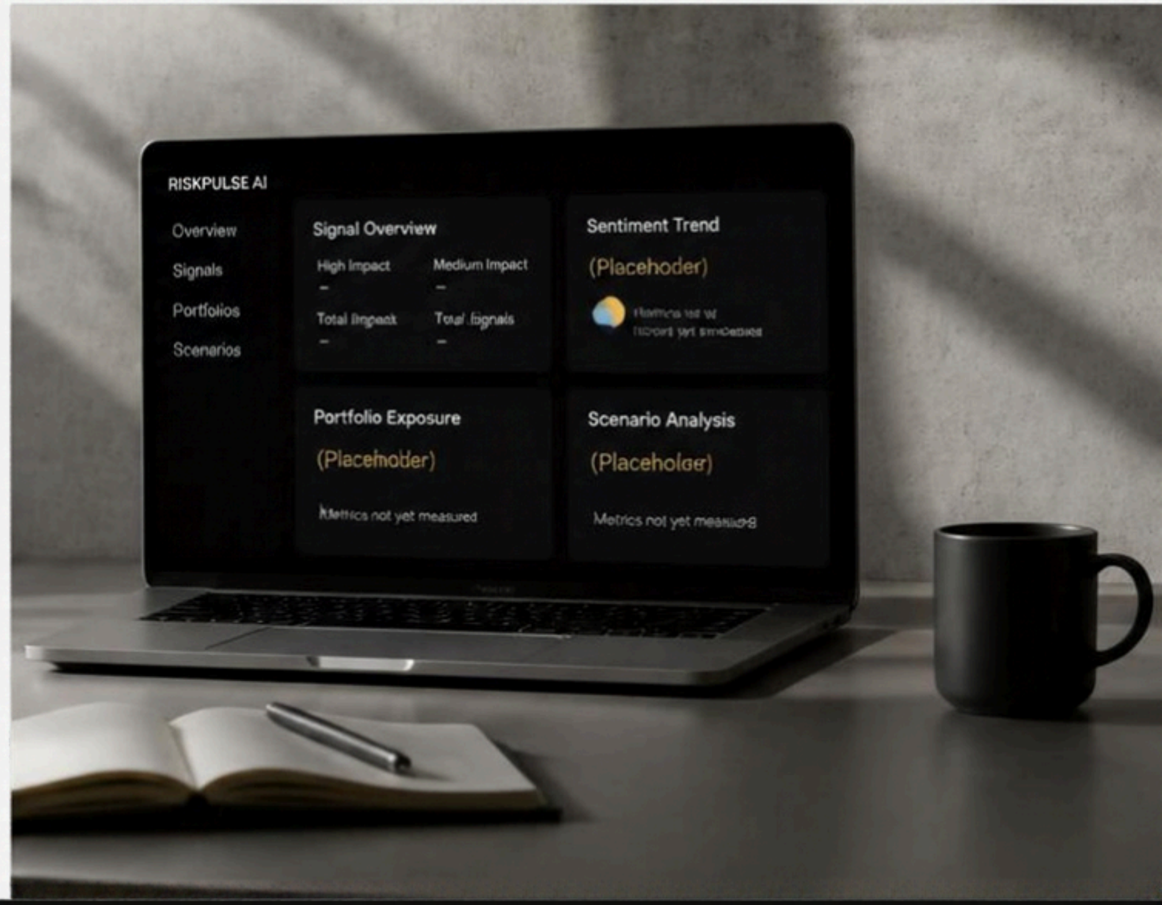
Decision support, not automated action

VALUE FOR PORTFOLIO RESEARCHERS

Richer event context for research and attribution

Better-informed scenario exploration and stress-testing

Stronger narrative around risk factors



not financial advice • not autonomous
results not guaranteed

Limitations, Roadmap & Conclusion

KEY LIMITATIONS

Relies on financial text streams that are high-volume and noisy

Sentiment alone does not reveal portfolio vulnerability

Simulation outputs reflect assumptions, not guaranteed outcomes

ROADMAP FOR VALIDATION & IMPROVEMENT

1 → Validate with public or synthetic data

2 → Build labelled solution datasets and measure quality and latency

3 → Improve entity resolution and scenario calibration

4 Strengthen attribution, monitoring, and auditenity

RiskPulse AI is an educational decision-support prototype.
It does not provide financial advice or autonomous trading.



CONCLUSION

RiskPulse AI connects financial text intelligence to explainable portfolio risk exploration.